

2nd Part Problem Set for Applied Abstract Algebra

Auslander–Reiten quivers

1. Let $M = (M_i, \varphi_\alpha) \in \text{rep } Q$ and $d_i = \dim M_i$. Show that

$$0 \longrightarrow M \longrightarrow \bigoplus_{i \in Q_0} d_i I(i) \longrightarrow \bigoplus_{\alpha: i \rightarrow j \in Q_1} d_j I(i) \longrightarrow 0$$

is an injective resolution for M .

Proof. Write

$$I^0 = \bigoplus_{i \in Q_0} M_i \otimes_k I(i), \quad I^1 = \bigoplus_{\alpha: i \rightarrow j} M_j \otimes_k I(i).$$

Since $\dim M_i = d_i$, these are isomorphic to the two direct sums in the statement. Both I^0 and I^1 are injective because they are finite direct sums of indecomposable injective representations.

For a vertex x , the space $I(i)_x$ has basis the paths $p : x \rightarrow i$. Hence

$$I_x^0 = \bigoplus_{p: x \rightarrow i} M_i, \quad I_x^1 = \bigoplus_{\alpha: i \rightarrow j, p: x \rightarrow i} M_j.$$

Define $\epsilon : M \rightarrow I^0$ by

$$\epsilon_x(m) = (M_p(m))_{p: x \rightarrow i},$$

where M_p denotes the composition of the arrow maps of M along the path p . This is a morphism of representations, and it is injective because the component indexed by the lazy path $e_x : x \rightarrow x$ is m itself.

Define $\delta : I^0 \rightarrow I^1$ at a vertex x by

$$(\delta_x((m_p)_{p: x \rightarrow i}))_{(\alpha: i \rightarrow j, p: x \rightarrow i)} = \varphi_\alpha(m_p) - m_{\alpha p},$$

where $\alpha p : x \rightarrow j$ is the path obtained by following p and then α . A direct check on paths shows that δ commutes with the arrow maps of the injectives, so δ is a morphism. Moreover,

$$\delta_x \epsilon_x(m)_{(\alpha, p)} = \varphi_\alpha(M_p(m)) - M_{\alpha p}(m) = 0,$$

so $\delta \epsilon = 0$.

If $(m_p) \in \ker \delta_x$, then for every path $p : x \rightarrow i$ one obtains inductively

$$m_p = M_p(m_{e_x}).$$

Thus $(m_p) = \epsilon_x(m_{e_x})$, and hence $\ker \delta = \text{im } \epsilon$. Finally, δ_x is surjective. Given $(y_{\alpha,p}) \in I_x^1$, define $m_{e_x} = 0$. Then, inductively on the length of $q = \alpha p$, set

$$m_{\alpha p} = \varphi_\alpha(m_p) - y_{\alpha,p}.$$

Every nontrivial path has a unique decomposition as a shorter path followed by its last arrow, so this definition is unambiguous. Since Q is acyclic, the induction terminates, and $\delta_x((m_p)) = (y_{\alpha,p})$. Therefore

$$0 \rightarrow M \xrightarrow{\epsilon} I^0 \xrightarrow{\delta} I^1 \rightarrow 0$$

is exact. Hence it is an injective resolution of M . □

2. Let $M \in \text{rep } Q$ and

$$0 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

be the standard projective resolution. Show that

$$0 \longrightarrow D(M) \longrightarrow D(P_0) \longrightarrow D(P_1) \longrightarrow 0$$

is the injective resolution for $D(M) \in \text{rep } Q^{\text{op}}$ defined in Exercise 1. Show that the duality functor D takes a minimal projective resolution for M to an injective envelope for $D(M)$.

Proof. The standard projective resolution has the form

$$0 \longrightarrow \bigoplus_{\alpha: i \rightarrow j} d_i P_Q(j) \longrightarrow \bigoplus_{i \in Q_0} d_i P_Q(i) \longrightarrow M \longrightarrow 0.$$

The duality $D = \text{Hom}_k(-, k)$ is exact and contravariant. Applying D gives

$$0 \longrightarrow D(M) \longrightarrow \bigoplus_{i \in Q_0} d_i D(P_Q(i)) \longrightarrow \bigoplus_{\alpha: i \rightarrow j} d_i D(P_Q(j)) \longrightarrow 0.$$

For every vertex i one has

$$D(P_Q(i)) \cong I_{Q^{\text{op}}}(i).$$

Also, an arrow $\alpha : i \rightarrow j$ in Q becomes an arrow $\alpha^{\text{op}} : j \rightarrow i$ in Q^{op} . Therefore the last term becomes

$$\bigoplus_{\alpha^{\text{op}}: j \rightarrow i} d_i I_{Q^{\text{op}}}(j),$$

which is exactly the injective resolution from Exercise 1 applied to $D(M)$.

Now suppose

$$0 \rightarrow P_1 \rightarrow P_0 \xrightarrow{\pi} M \rightarrow 0$$

is minimal, so $\pi : P_0 \rightarrow M$ is a projective cover. Applying D gives a monomorphism

$$D(M) \xrightarrow{D\pi} D(P_0).$$

Since D sends projectives to injectives and epimorphisms to monomorphisms, $D(P_0)$ is injective. Minimality of the projective cover is dual to minimality of the injective envelope: if $D(M) \hookrightarrow J$ is another monomorphism into an injective representation, dualizing gives an epimorphism $D(J) \rightarrow M$ from a projective representation, and the defining property of the projective cover gives a surjection $D(J) \twoheadrightarrow P_0$. Dualizing back gives a monomorphism $D(P_0) \hookrightarrow J$ extending $D(M) \hookrightarrow J$. Hence

$$D(M) \hookrightarrow D(P_0)$$

is an injective envelope. Moreover, if $P_1 \rightarrow \ker \pi$ is the projective cover in the minimal projective resolution, then dualizing it shows that

$$\operatorname{coker}(D\pi) \cong D(\ker \pi) \longrightarrow D(P_1)$$

is an injective envelope. Hence the dual sequence is the minimal injective resolution of $D(M)$. $\square$

3. Let Q be the Kronecker quiver

$$1 \begin{matrix} \xleftarrow{\beta} \\ \xleftarrow{\alpha} \end{matrix} 2$$

For $\lambda \in k \cup \{\infty\}$, let M_λ be given by

$$M_\lambda : \quad k \xleftarrow[\lambda]{1} k \quad (\lambda \in k), \quad M_\infty : \quad k \xleftarrow[1]{0} k.$$

Recall that $S(1) : k \leftarrow 0$ and $S(2) : 0 \leftarrow k$.

(a) Show that the short exact sequences

$$0 \rightarrow S(1) \rightarrow M_\lambda \rightarrow S(2) \rightarrow 0, \quad 0 \rightarrow S(1) \rightarrow M_\mu \rightarrow S(2) \rightarrow 0$$

are not equivalent if $\lambda \neq \mu$.

Proof. In the sequence

$$0 \rightarrow S(1) \rightarrow M_\lambda \rightarrow S(2) \rightarrow 0,$$

the inclusion is the identity at vertex 1 and zero at vertex 2, while the projection is zero at vertex 1 and the identity at vertex 2. If two such extensions were equivalent, then there would be an isomorphism $f = (f_1, f_2) : M_\lambda \rightarrow M_\mu$ inducing the identity on both $S(1)$ and $S(2)$. Thus $f_1 = 1$ and $f_2 = 1$. For $\lambda, \mu \in k$, commutativity along the two arrows gives

$$1 \cdot 1 = 1 \cdot 1, \quad 1 \cdot \lambda = \mu \cdot 1.$$

Hence $\lambda = \mu$. The cases involving ∞ are identical: the pair of arrow maps $(0, 1)$ cannot be carried to $(1, \lambda)$ by an equivalence fixing the end terms. Therefore the two extensions are not equivalent when $\lambda \neq \mu$. $\square$

(b) Show that $\tau M_\lambda = M_\lambda$ for all λ .

Proof. We compute the Auslander–Reiten translate using a minimal projective resolution. The projective representation $P(1)$ is $S(1)$, and

$$P(2) : \quad k^2 \xleftarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}} k.$$

For $\lambda \in k$, the projective cover $P(2) \rightarrow M_\lambda$ is the identity at vertex 2 and the row $(1 \ \lambda) : k^2 \rightarrow k$ at vertex 1. Its kernel is $P(1)$, embedded at vertex 1 by the vector $(-\lambda, 1)^T$. Hence

$$0 \rightarrow P(1) \xrightarrow{u_\lambda} P(2) \rightarrow M_\lambda \rightarrow 0$$

is a minimal projective resolution. Applying the Nakayama functor gives

$$0 \rightarrow \tau M_\lambda \rightarrow I(1) \xrightarrow{\nu u_\lambda} I(2).$$

Now $I(1)$ has vector spaces k at vertex 1 and k^2 at vertex 2, with arrow maps $(1 \ 0)$ and $(0 \ 1)$. The map νu_λ at vertex 2 is the row $(-\lambda \ 1)$, so its kernel is the line spanned by $(1, \lambda)^T$. Restricted to this line, the two arrow maps are 1 and λ . Thus

$$\tau M_\lambda \cong M_\lambda.$$

For $\lambda = \infty$, the same computation uses the embedding $P(1) \rightarrow P(2)$ by $(1, 0)^T$; the kernel of the dual row is spanned by $(0, 1)^T$, giving arrow maps $(0, 1)$, hence $\tau M_\infty \cong M_\infty$. $\square$

4. Prove that P is projective if and only if $\text{Ext}^1(P, N) = 0$ for all representations N .

Proof. If P is projective, then $\text{Hom}(P, -)$ is exact. Therefore applying $\text{Hom}(P, -)$ to any short exact sequence produces an exact sequence without a nonzero connecting obstruction, so $\text{Ext}^1(P, N) = 0$ for all N .

Conversely, assume $\text{Ext}^1(P, N) = 0$ for all N . Let $g : M \twoheadrightarrow X$ be an epimorphism and let $f : P \rightarrow X$ be any morphism. Form the pullback

$$E = M \times_X P = \{(m, p) : g(m) = f(p)\}.$$

Then there is a short exact sequence

$$0 \rightarrow \ker g \rightarrow E \rightarrow P \rightarrow 0.$$

Its extension class lies in $\text{Ext}^1(P, \ker g)$, which is zero by assumption. Hence the sequence splits. Therefore there exists a section $s : P \rightarrow E$. Composing s with the projection $E \rightarrow M$ gives a morphism $\tilde{f} : P \rightarrow M$ satisfying $g\tilde{f} = f$. Thus every morphism from P lifts across every epimorphism, so P is projective. $\square$

5. Show that isoclasses of indecomposable representations of quivers of type A_n are determined by their dimension vectors of the form $(0, \dots, 0, 1, \dots, 1, 0, \dots, 0)$, and the corresponding representation is $M = (M_i, \varphi_\alpha)$ with $M_i = k$ if the dimension at i is one

and $M_i = 0$ otherwise; and $\varphi_\alpha = 1$ if the dimension at $s(\alpha)$ and at $t(\alpha)$ is one, and $\varphi_\alpha = 0$ otherwise.

Proof. Let Q have underlying graph

$$1 - 2 - \cdots - n.$$

For $1 \leq a \leq b \leq n$, define $M[a, b]$ by

$$M[a, b]_i = \begin{cases} k, & a \leq i \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

and each arrow map inside the interval $[a, b]$ equal to the identity, while all other arrow maps are zero. The support of $M[a, b]$ is connected, and all nonzero vertex spaces are one-dimensional. Hence a nontrivial direct-sum decomposition would split a connected string of identity maps into two disjoint nonzero subrepresentations, which is impossible. Thus $M[a, b]$ is indecomposable.

Conversely, by Gabriel's theorem for Dynkin quivers, the dimension vectors of indecomposable representations are precisely the positive roots of type A_n . These positive roots are exactly the interval vectors

$$(0, \dots, 0, \underbrace{1, \dots, 1}_{a \leq i \leq b}, 0, \dots, 0).$$

For each such vector the representation $M[a, b]$ defined above is indecomposable, so every indecomposable is isomorphic to one of the $M[a, b]$. The dimension vector of $M[a, b]$ is precisely

$$(0, \dots, 0, \underbrace{1, \dots, 1}_{a \leq i \leq b}, 0, \dots, 0),$$

and the representation is uniquely determined by this vector because the only nonzero maps in the connected support can be scaled to the identity. This proves the claim. $\square$

6. Compute the Auslander–Reiten quiver for

$$1 \longleftarrow 2 \longleftarrow 3 \longrightarrow 4 \longleftarrow 5$$

Proof. Write $\mathbf{d}_1 \mathbf{d}_2 \mathbf{d}_3 \mathbf{d}_4 \mathbf{d}_5$ for the indecomposable representation with dimension vector $(d_1, d_2, d_3, d_4, d_5)$. The τ^{-1} -orbits are

$$\begin{aligned} 10000 &\rightarrow 01000 \rightarrow 00110 \rightarrow 00001, \\ 11000 &\rightarrow 01110 \rightarrow 00111, \\ 11110 &\rightarrow 01111 \rightarrow 00100, \\ 00010 &\rightarrow 11111 \rightarrow 01100, \\ 00011 &\rightarrow 11100. \end{aligned}$$

The irreducible arrows are

$$\begin{aligned}
& \mathbf{00010} \rightarrow \mathbf{00011}, \quad \mathbf{00010} \rightarrow \mathbf{11110}, \quad \mathbf{00011} \rightarrow \mathbf{11111}, \quad \mathbf{00110} \rightarrow \mathbf{00111}, \\
& \mathbf{00111} \rightarrow \mathbf{00001}, \quad \mathbf{00111} \rightarrow \mathbf{00100}, \quad \mathbf{01000} \rightarrow \mathbf{01110}, \quad \mathbf{01100} \rightarrow \mathbf{00100}, \\
& \mathbf{01110} \rightarrow \mathbf{00110}, \quad \mathbf{01110} \rightarrow \mathbf{01111}, \quad \mathbf{01111} \rightarrow \mathbf{00111}, \quad \mathbf{01111} \rightarrow \mathbf{01100}, \\
& \mathbf{10000} \rightarrow \mathbf{11000}, \quad \mathbf{11000} \rightarrow \mathbf{01000}, \quad \mathbf{11000} \rightarrow \mathbf{11110}, \quad \mathbf{11100} \rightarrow \mathbf{01100}, \\
& \mathbf{11110} \rightarrow \mathbf{01110}, \quad \mathbf{11110} \rightarrow \mathbf{11111}, \quad \mathbf{11111} \rightarrow \mathbf{01111}, \quad \mathbf{11111} \rightarrow \mathbf{11100}.
\end{aligned}$$

Together with the above vertices, these arrows give the Auslander–Reiten quiver. $\square$

7. Compute the Auslander–Reiten quiver for

$$1 \longrightarrow 2 \longleftarrow 3 \longrightarrow 4 \longleftarrow 5 \longleftarrow 6$$

Proof. Write $\mathbf{d}_1 \cdots \mathbf{d}_6$ for the indecomposable with dimension vector $(d_1, \dots, d_6)$. The τ^{-1} -orbits are

$$\begin{aligned}
& \mathbf{110000} \rightarrow \mathbf{001100} \rightarrow \mathbf{000010} \rightarrow \mathbf{000001}, \\
& \mathbf{010000} \rightarrow \mathbf{111100} \rightarrow \mathbf{001110} \rightarrow \mathbf{000011}, \\
& \quad \mathbf{011100} \rightarrow \mathbf{111110} \rightarrow \mathbf{001111}, \\
& \mathbf{000100} \rightarrow \mathbf{011110} \rightarrow \mathbf{111111} \rightarrow \mathbf{001000}, \\
& \quad \mathbf{000110} \rightarrow \mathbf{011111} \rightarrow \mathbf{111000}, \\
& \quad \mathbf{000111} \rightarrow \mathbf{011000} \rightarrow \mathbf{100000}.
\end{aligned}$$

The irreducible arrows are

$$\begin{aligned}
& \mathbf{000010} \rightarrow \mathbf{000011}, \quad \mathbf{000011} \rightarrow \mathbf{000001}, \quad \mathbf{000100} \rightarrow \mathbf{000110}, \\
& \mathbf{000100} \rightarrow \mathbf{011100}, \quad \mathbf{000110} \rightarrow \mathbf{000111}, \quad \mathbf{000110} \rightarrow \mathbf{011110}, \\
& \mathbf{000111} \rightarrow \mathbf{011111}, \quad \mathbf{001100} \rightarrow \mathbf{001110}, \quad \mathbf{001110} \rightarrow \mathbf{000010}, \\
& \mathbf{001110} \rightarrow \mathbf{001111}, \quad \mathbf{001111} \rightarrow \mathbf{000011}, \quad \mathbf{001111} \rightarrow \mathbf{001000}, \\
& \mathbf{010000} \rightarrow \mathbf{011100}, \quad \mathbf{010000} \rightarrow \mathbf{110000}, \quad \mathbf{011000} \rightarrow \mathbf{111000}, \\
& \mathbf{011100} \rightarrow \mathbf{011110}, \quad \mathbf{011100} \rightarrow \mathbf{111100}, \quad \mathbf{011110} \rightarrow \mathbf{011111}, \\
& \mathbf{011110} \rightarrow \mathbf{111110}, \quad \mathbf{011111} \rightarrow \mathbf{011000}, \quad \mathbf{011111} \rightarrow \mathbf{111111}, \\
& \mathbf{110000} \rightarrow \mathbf{111100}, \quad \mathbf{111000} \rightarrow \mathbf{001000}, \quad \mathbf{111000} \rightarrow \mathbf{100000}, \\
& \mathbf{111100} \rightarrow \mathbf{001100}, \quad \mathbf{111100} \rightarrow \mathbf{111110}, \quad \mathbf{111110} \rightarrow \mathbf{001110}, \\
& \mathbf{111110} \rightarrow \mathbf{111111}, \quad \mathbf{111111} \rightarrow \mathbf{001111}, \quad \mathbf{111111} \rightarrow \mathbf{111000}.
\end{aligned}$$

This determines the Auslander–Reiten quiver. $\square$

8. The Cartan matrix $C = (c_{ij})_{1 \leq i, j \leq n}$ of the quiver Q , where c_{ij} is the number of paths from j to i and n is the number of vertices in Q . Show that C is invertible and find C^{-1} .

Proof. Let $A = (a_{ij})$ be the arrow matrix, where a_{ij} is the number of arrows from j to i . Since Q has no oriented cycles, A is nilpotent after a suitable ordering of vertices. The

matrix A^m counts paths of length m . Hence the Cartan matrix is

$$C = I + A + A^2 + \cdots + A^N$$

for N sufficiently large. Since A is nilpotent,

$$(I - A)(I + A + A^2 + \cdots + A^N) = I.$$

Thus

$$C^{-1} = I - A.$$

Equivalently,

$$(C^{-1})_{ij} = \begin{cases} 1, & i = j, \\ -\#\{\text{arrows } j \rightarrow i\}, & i \neq j. \end{cases}$$

In particular, C is invertible. □

9. Let Q be a quiver of type A with n vertices and consider the triangulation corresponding to Q . Show that if we consider an elementary clockwise rotation of the diagonal corresponding to 1, we obtain the diagonal corresponding to $P(1)$. Show that if we consider an elementary counterclockwise rotation of the diagonal corresponding to 1, we obtain the diagonal corresponding to $I(1)$.

Proof. In the polygon model for type A_n , an indecomposable representation is represented by a diagonal γ , and the i -th coordinate of its dimension vector is the crossing number of γ with the diagonal labeled i in the chosen triangulation. We use the rotation convention of this exercise: its elementary clockwise rotation is Schiffler's τ^{-1} -rotation. With Schiffler's convention, the words "clockwise" and "counterclockwise" in the statement are interchanged.

Rotate the diagonal labeled 1 clockwise by one elementary step. By the construction of the triangulation from the quiver, the new diagonal crosses exactly the diagonals labeled j for which there exists an oriented path

$$1 \rightsquigarrow j$$

in Q . Therefore its dimension vector is

$$(\#\{\text{paths } 1 \rightarrow j\})_{j \in Q_0},$$

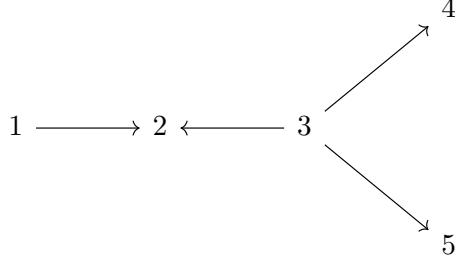
which is precisely the dimension vector of the projective representation $P(1)$. Since type A indecomposables are determined by their dimension vectors, this rotated diagonal corresponds to $P(1)$.

Similarly, rotating the diagonal labeled 1 counterclockwise crosses exactly the diagonals j for which there exists an oriented path

$$j \rightsquigarrow 1.$$

This is the dimension vector of the injective representation $I(1)$. Hence the counter-clockwise rotation gives the diagonal corresponding to $I(1)$. $\square$

10. Compute the Auslander–Reiten quiver for



Proof. Write $\mathbf{d}_1\mathbf{d}_2\mathbf{d}_3\mathbf{d}_4\mathbf{d}_5$ for the indecomposable representation with dimension vector $(d_1, d_2, d_3, d_4, d_5)$. The τ^{-1} -orbits are

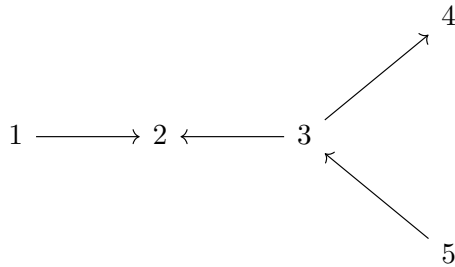
$$\begin{aligned}
& \mathbf{11000} \rightarrow \mathbf{00111} \rightarrow \mathbf{01100} \rightarrow \mathbf{10000}, \\
& \mathbf{01000} \rightarrow \mathbf{11111} \rightarrow \mathbf{01211} \rightarrow \mathbf{11100}, \\
& \mathbf{01111} \rightarrow \mathbf{12211} \rightarrow \mathbf{11211} \rightarrow \mathbf{00100}, \\
& \mathbf{00010} \rightarrow \mathbf{01101} \rightarrow \mathbf{11110} \rightarrow \mathbf{00101}, \\
& \mathbf{00001} \rightarrow \mathbf{01110} \rightarrow \mathbf{11101} \rightarrow \mathbf{00110}.
\end{aligned}$$

The irreducible arrows are

$$\begin{aligned}
& \mathbf{00001} \rightarrow \mathbf{01111}, & \mathbf{00010} \rightarrow \mathbf{01111}, & \mathbf{00101} \rightarrow \mathbf{00100}, & \mathbf{00110} \rightarrow \mathbf{00100}, \\
& \mathbf{00111} \rightarrow \mathbf{01211}, & \mathbf{01000} \rightarrow \mathbf{01111}, & \mathbf{01000} \rightarrow \mathbf{11000}, & \mathbf{01100} \rightarrow \mathbf{11100}, \\
& \mathbf{01101} \rightarrow \mathbf{12211}, & \mathbf{01110} \rightarrow \mathbf{12211}, & \mathbf{01111} \rightarrow \mathbf{01101}, & \mathbf{01111} \rightarrow \mathbf{01110}, \\
& \mathbf{01111} \rightarrow \mathbf{11111}, & \mathbf{01211} \rightarrow \mathbf{01100}, & \mathbf{01211} \rightarrow \mathbf{11211}, & \mathbf{11000} \rightarrow \mathbf{11111}, \\
& \mathbf{11100} \rightarrow \mathbf{00100}, & \mathbf{11100} \rightarrow \mathbf{10000}, & \mathbf{11101} \rightarrow \mathbf{11211}, & \mathbf{11110} \rightarrow \mathbf{11211}, \\
& \mathbf{11111} \rightarrow \mathbf{00111}, & \mathbf{11111} \rightarrow \mathbf{12211}, & \mathbf{11211} \rightarrow \mathbf{00101}, & \mathbf{11211} \rightarrow \mathbf{00110}, \\
& \mathbf{11211} \rightarrow \mathbf{11100}, & \mathbf{12211} \rightarrow \mathbf{01211}, & \mathbf{12211} \rightarrow \mathbf{11101}, & \mathbf{12211} \rightarrow \mathbf{11110}.
\end{aligned}$$

These vertices and arrows give the Auslander–Reiten quiver. $\square$

11. Compute the Auslander–Reiten quiver for



Proof. Again write $\mathbf{d}_1\mathbf{d}_2\mathbf{d}_3\mathbf{d}_4\mathbf{d}_5$ for the dimension vector. The τ^{-1} -orbits are

$$\begin{aligned} & \mathbf{11000} \rightarrow \mathbf{00110} \rightarrow \mathbf{01101} \rightarrow \mathbf{10000}, \\ & \mathbf{01000} \rightarrow \mathbf{11110} \rightarrow \mathbf{01211} \rightarrow \mathbf{11101}, \\ & \mathbf{01110} \rightarrow \mathbf{12211} \rightarrow \mathbf{11211} \rightarrow \mathbf{00101}, \\ & \mathbf{00010} \rightarrow \mathbf{01100} \rightarrow \mathbf{11111} \rightarrow \mathbf{00100} \rightarrow \mathbf{00001}, \\ & \mathbf{01111} \rightarrow \mathbf{11100} \rightarrow \mathbf{00111}. \end{aligned}$$

The irreducible arrows are

$$\begin{aligned} & \mathbf{00010} \rightarrow \mathbf{01110}, \quad \mathbf{00100} \rightarrow \mathbf{00101}, \quad \mathbf{00101} \rightarrow \mathbf{00001}, \quad \mathbf{00110} \rightarrow \mathbf{01211}, \\ & \mathbf{00111} \rightarrow \mathbf{00101}, \quad \mathbf{01000} \rightarrow \mathbf{01110}, \quad \mathbf{01000} \rightarrow \mathbf{11000}, \quad \mathbf{01100} \rightarrow \mathbf{12211}, \\ & \mathbf{01101} \rightarrow \mathbf{11101}, \quad \mathbf{01110} \rightarrow \mathbf{01100}, \quad \mathbf{01110} \rightarrow \mathbf{01111}, \quad \mathbf{01110} \rightarrow \mathbf{11110}, \\ & \mathbf{01111} \rightarrow \mathbf{12211}, \quad \mathbf{01211} \rightarrow \mathbf{01101}, \quad \mathbf{01211} \rightarrow \mathbf{11211}, \quad \mathbf{11000} \rightarrow \mathbf{11110}, \\ & \mathbf{11100} \rightarrow \mathbf{11211}, \quad \mathbf{11101} \rightarrow \mathbf{00101}, \quad \mathbf{11101} \rightarrow \mathbf{10000}, \quad \mathbf{11110} \rightarrow \mathbf{00110}, \\ & \mathbf{11110} \rightarrow \mathbf{12211}, \quad \mathbf{11111} \rightarrow \mathbf{11211}, \quad \mathbf{11211} \rightarrow \mathbf{00100}, \quad \mathbf{11211} \rightarrow \mathbf{00111}, \\ & \mathbf{11211} \rightarrow \mathbf{11101}, \quad \mathbf{12211} \rightarrow \mathbf{01211}, \quad \mathbf{12211} \rightarrow \mathbf{11100}, \quad \mathbf{12211} \rightarrow \mathbf{11111}. \end{aligned}$$

These data determine the Auslander–Reiten quiver. □

12. Let Q be the quiver from Problem 10 and consider the indecomposable representations

$$\underline{\dim} L = (0, 1, 1, 1, 0), \quad \underline{\dim} N = (1, 1, 2, 1, 1).$$

Prove that there is a unique representation M for which there exists a non-split short exact sequence

$$0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0.$$

Proof. From the Auslander–Reiten quiver in Problem 10, $N = \mathbf{11211}$ and

$$\tau N = \mathbf{12211}.$$

Using the Auslander–Reiten formula,

$$\mathrm{Ext}^1(N, L) \cong D \mathrm{Hom}(L, \tau N).$$

The type D hammock starting at $L = \mathbf{01110}$ labels $\tau N = \mathbf{12211}$ with multiplicity 1; equivalently, the mesh recursion gives exactly one contribution from the sectional arrow

$$\mathbf{01110} \longrightarrow \mathbf{12211}.$$

Hence

$$\dim \mathrm{Hom}(L, \tau N) = 1.$$

Therefore

$$\dim \mathrm{Ext}^1(N, L) = 1.$$

Thus there is, up to scalar, a unique nonzero extension class. Since indecomposable representations of Dynkin quivers are bricks, scalar automorphisms of L and N identify the nonzero scalar multiples of this class. Consequently all non-split extensions have isomorphic middle term.

The middle term is read from the two sectional boundary paths in the Auslander–Reiten quiver, or equivalently by adding the corresponding middle vertices in the mesh calculation. One obtains

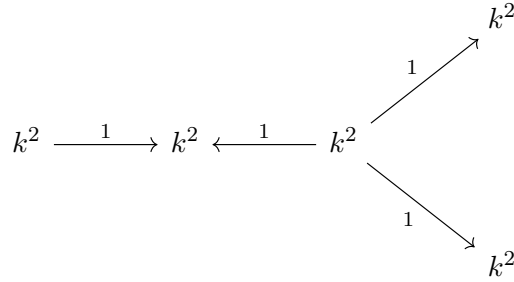
$$M \cong \mathbf{01211} \oplus \mathbf{11110}.$$

Indeed,

$$\underline{\dim}(\mathbf{01211} \oplus \mathbf{11110}) = (0, 1, 2, 1, 1) + (1, 1, 1, 1, 0) = (1, 2, 3, 2, 1) = \underline{\dim}L + \underline{\dim}N.$$

It represents the unique nonzero class in $\text{Ext}^1(N, L)$. Thus this M is unique up to isomorphism. $\square$

- 13.** Decompose the following representation of Q from Problem 10 into a direct sum of indecomposable representations:



all maps are identities.

Proof. Let X be the indecomposable representation with dimension vector

$$\underline{\dim}X = (1, 1, 1, 1, 1)$$

and with all nonzero arrow maps equal to the identity $k \rightarrow k$. The given representation has k^2 at every vertex and each arrow map is the identity on k^2 . Choose a basis e_1, e_2 of k^2 at every vertex. The subspaces spanned by e_1 at all vertices form a subrepresentation isomorphic to X , and the subspaces spanned by e_2 at all vertices form another subrepresentation isomorphic to X . Since the identity maps preserve this decomposition, we get

$$M \cong X \oplus X.$$

Thus the required decomposition is

$$M \cong \mathbf{11111} \oplus \mathbf{11111}.$$

$\square$

14. Let Q be a quiver of type D with n vertices and consider the triangulation corresponding to Q . Show that if we consider an elementary clockwise rotation plus tagging of the diagonals corresponding to $n-1$ and n , we obtain the diagonals corresponding to $P(n-1)$ and $P(n)$ respectively.

Proof. In the punctured polygon model for type D_n , the two vertices $n-1$ and n correspond to the two tagged arcs incident with the puncture. The dimension vector of the representation associated with an arc is given by its crossing numbers with the arcs of the fixed tagged triangulation. We use the rotation convention of this exercise: its elementary clockwise rotation is Schiffler's τ^{-1} -rotation. With Schiffler's convention, the words "clockwise" and "counterclockwise" in the statement are interchanged.

For an ordinary vertex i , the same argument as in type A says that an elementary clockwise rotation of the corresponding arc crosses exactly the arcs labeled by vertices reachable from i by oriented paths. For the puncture arcs, the elementary rotation must be accompanied by changing the tag; otherwise the resulting arc has the wrong tagged intersection with the puncture arcs. Thus the operation "clockwise rotation plus tagging" gives an arc whose crossing vector is

$$(\#\{\text{paths } n-1 \rightarrow j\})_{j \in Q_0}$$

for the arc labeled $n-1$, and similarly

$$(\#\{\text{paths } n \rightarrow j\})_{j \in Q_0}$$

for the arc labeled n . These are exactly the dimension vectors of $P(n-1)$ and $P(n)$. Since indecomposables of Dynkin type D are determined by their positive-root dimension vectors, the resulting tagged arcs correspond to $P(n-1)$ and $P(n)$ respectively. $\square$

15. Construct the decomposition of E_d into orbits for

$$Q : \quad 1 \longrightarrow 2 \longleftarrow 3, \quad d = (2, 3, 1).$$

Proof. A representation is a pair

$$A : k^2 \rightarrow k^3, \quad b : k \rightarrow k^3.$$

The group $G_d = \text{GL}_2 \times \text{GL}_3 \times \text{GL}_1$ acts by base change. The orbit is determined by

$$r = \text{rank } A, \quad s = \text{rank } b, \quad u = \dim(\text{im } A + \text{im } b).$$

Equivalently, by Krull-Schmidt, the orbit is determined by the multiplicities of the indecomposables

$$S(1), S(2), S(3), P(1), P(3), I(2).$$

Solving

$$x_1 S(1) + x_2 S(2) + x_3 S(3) + x_4 P(1) + x_5 P(3) + x_6 I(2)$$

with dimension vector $(2, 3, 1)$ gives the following eight orbits:

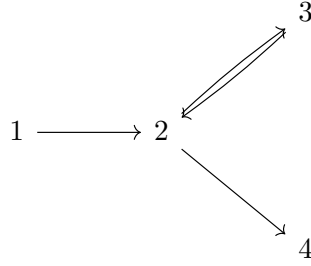
(r, s, u)	decomposition
$(2, 1, 3)$	$P(1)^2 \oplus P(3)$
$(2, 1, 2)$	$P(1) \oplus I(2) \oplus S(2)$
$(2, 0, 2)$	$P(1)^2 \oplus S(2) \oplus S(3)$
$(1, 1, 2)$	$S(1) \oplus S(2) \oplus P(1) \oplus P(3)$
$(1, 1, 1)$	$S(1) \oplus S(2)^2 \oplus I(2)$
$(1, 0, 1)$	$S(1) \oplus S(2)^2 \oplus S(3) \oplus P(1)$
$(0, 1, 1)$	$S(1)^2 \oplus S(2)^2 \oplus P(3)$
$(0, 0, 0)$	$S(1)^2 \oplus S(2)^3 \oplus S(3)$.

Thus

$$E_d = \bigsqcup_{(r,s,u)} \mathcal{O}_{r,s,u}$$

with the eight orbits listed above. □

16. Construct the decomposition of E_d into orbits for $d = (2, 2, 1, 1)$ and



Proof. Let the arrows be

$$a : 1 \rightarrow 2, \quad b : 2 \rightarrow 3, \quad c : 3 \rightarrow 2, \quad d : 2 \rightarrow 4.$$

A point of E_d is a quadruple

$$(A, B, C, D) \in \text{Mat}_{2 \times 2}(k) \oplus \text{Mat}_{1 \times 2}(k) \oplus \text{Mat}_{2 \times 1}(k) \oplus \text{Mat}_{1 \times 2}(k).$$

Because the quiver contains the oriented two-cycle $2 \rightleftarrows 3$, the orbit decomposition is not finite. Indeed, consider the family

$$A = 0, \quad D = 0, \quad C = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad B_\lambda = \begin{pmatrix} \lambda & 0 \end{pmatrix}, \quad \lambda \in k.$$

For such a representation,

$$B_\lambda C = \lambda \in \text{End}(k).$$

Under base change, B and C transform as

$$B \mapsto g_3 B g_2^{-1}, \quad C \mapsto g_2 C g_3^{-1},$$

so the scalar BC is invariant:

$$(g_3 B g_2^{-1})(g_2 C g_3^{-1}) = g_3(BC)g_3^{-1} = BC.$$

Therefore the representations corresponding to distinct λ lie in distinct G_d -orbits. Hence E_d contains the disjoint union

$$\bigsqcup_{\lambda \in k} \mathcal{O}_\lambda,$$

where $\mathcal{O}_\lambda$ is the orbit of $(0, B_\lambda, C, 0)$. In particular, unlike the Dynkin cases, E_d has infinitely many orbits. Thus no finite orbit decomposition exists; the orbit decomposition is an infinite one, already containing the one-parameter family above. $\square$

17. Let $q : \mathbb{Z}^n \rightarrow \mathbb{Z}$ be a quadratic form. Show that

$$(x, y) = q(x + y) - q(x) - q(y)$$

is symmetric and bilinear.

Proof. Write

$$q(x) = \sum_i a_i x_i^2 + \sum_{i < j} a_{ij} x_i x_j.$$

Then

$$q(x + y) - q(x) - q(y) = \sum_i 2a_i x_i y_i + \sum_{i < j} a_{ij} (x_i y_j + x_j y_i).$$

This expression is linear in x for fixed y and linear in y for fixed x , so it is bilinear. It is also unchanged when x and y are interchanged. Hence the form is symmetric and bilinear. $\square$

18. Prove that for $n \leq 5$ any tree quiver is Dynkin or Euclidean.

Proof. The representation type depends only on the underlying graph. If a tree with at most five vertices has no branching vertex, then it is a path, hence of Dynkin type A_n . If it has a branching vertex, then for $n \leq 5$ the only possibilities are:

$$D_4, \quad D_5, \quad \tilde{D}_4.$$

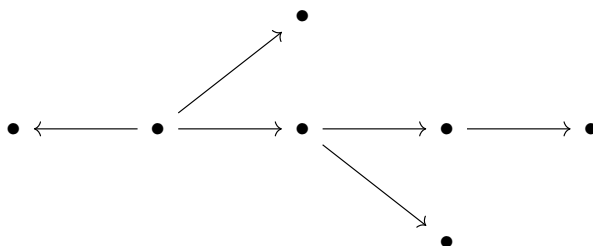
Here D_4 and D_5 are Dynkin, and $\tilde{D}_4$ is Euclidean. Therefore every tree quiver with $n \leq 5$ is Dynkin or Euclidean. $\square$

19. Show that the smallest wild tree with valence at most 3 at each vertex has 7 vertices.

Proof. By Problem 18, no tree with at most five vertices is wild. For six vertices and maximum valence at most 3, the possible underlying trees are

$$A_6, \quad D_6, \quad E_6, \quad \tilde{D}_5.$$

Now consider the seven-vertex tree



Show that the smallest wild tree that has at most one vertex of valence 3 has 8 vertices.

$$a \leq b \leq c.$$
$$(1, 1, c), \quad (1, 2, 2), \quad (1, 2, 3), \quad (2, 2, 2).$$

For eight vertices, take the three-armed tree with arm lengths

$(2, 2, 3).$

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